Stochastic Processes

Lecture 19

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An Important Result

Lemma (An Important Result)

1. *For any state* $y \in \mathcal{X}$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} P^k(y,y) = \begin{cases} 0, & f_{yy} < 1, \\ \dfrac{1}{\mu_{yy} - 1}, & f_{yy} = 1. \end{cases}$$

2. *If* x, y *are two states such that* $f_{xy} = 1$, *then*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} P^k(x,y) = \begin{cases} 0, & f_{yy} < 1, \\ \dfrac{1}{\mu_{yy} - 1}, & f_{yy} = 1. \end{cases}$$

On Open and Closed Communicating Classes

Lemma (On Open and Closed Communicating Classes)

Every state in an open communicating class is transient.
If a communicating class is closed and has finitely many states, then every state in it is positive recurrent.

Corollary:

- Irreducible $+$ finite state space $\implies$ positive recurrence

Proof of Lemma 2
(Open Communicating Classes are Transient)

- Suppose $\mathcal{C} \subseteq \mathcal{X}$ is an open communicating class
  By definition, there exist $x \in \mathcal{C}$ and $y \in \mathcal{C}^{\complement}$ such that $P(x, y) > 0$
- We then have

$$
\begin{aligned}
f_{xx} &= \mathbb{P}(\{\tau_x^{(1)} < +\infty\} \mid \{X_1 = x\}) = \sum_{z \in \mathcal{X}} \mathbb{P}(\{\tau_x^{(1)} < +\infty, X_2 = z\} \mid \{X_1 = x\}) \\
&= P(x, x) + \sum_{z \neq x} P(x, z) f_{zx} = P(x, x) + P(x, y) f_{yx} + \sum_{z \notin \{x, y\}} P(x, z) f_{zx} \\
&= P(x, x) + \sum_{z \notin \{x, y\}} P(x, z) f_{zx} \\
&\leq P(x, x) + \sum_{z \notin \{x, y\}} P(x, z) \\
&< P(x, x) + \sum_{z \neq x} P(x, z) = 1.
\end{aligned}
$$

- The strict inequality above follows from the fact that $P(x, y) > 0$

Proof of Lemma 2
(Finite Closed Communicating Classes are Positive Recurrent)

- The previous result automatically implies that every state in a closed communicating class is recurrent
- Suppose $\mathcal{C} \subseteq \mathcal{X}$ is closed and has finitely many states. Fix $x \in \mathcal{C}$.
- Because $\mathcal{C}$ is closed, we have

$$\sum_{y \in \mathcal{C}} P^k(x, y) = 1 \qquad \forall\, k \in \mathbb{N}.$$

- Also, $f_{xy} = 1$ for every $y \in \mathcal{C}$, and therefore by Lemma 1,

$$\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} P^k(x, y) = \frac{1}{\mu_{yy} - 1} \qquad \forall\, y \in \mathcal{C}.$$

- If each state $y \in \mathcal{C}$ is null recurrent, then

$$0 = \sum_{y \in \mathcal{C}} \frac{1}{\mu_{yy} - 1} = \sum_{y \in \mathcal{C}} \lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} P^k(x, y) = \lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} \sum_{y \in \mathcal{C}} P^k(x, y) = 1 \qquad \textbf{(contradiction!)}$$

- Thus, every state in $\mathcal{C}$ must be positive recurrent

Invariant (Stationary) Distribution

Definition (Invariant Distribution)

Let $\mathbf{X} \overset{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P.
A PMF $\pi = [\pi_x : x \in \mathcal{X}]$ on $\mathcal{X}$ is called the **invariant distribution** for P if

$$\pi = \pi P. \qquad \textbf{(global balance equation)}.$$

That is, for all $y \in \mathcal{X}$,

$$\pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y).$$

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Remarks on Invariant Distribution

Invariant Distribution

$$\pi = \pi P, \qquad \pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y) \quad \forall y \in \mathcal{X}.$$

- Global balance equation is a collection of $|\mathcal{X}|$ linear equations
- π is a **non-negative, left eigenvector** of P with eigenvalue 1
- From Chapman–Kolmogorov's equations,

$$\pi = \pi P^n \qquad \text{for any } n \in \mathbb{N}.$$

- If $\mu = \pi$ (i.e., $X_1 \sim \pi$), then:
  - $X_n \sim \pi$ for all $n \in \mathbb{N}$
  - $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a stationary process (i.e., all FDDs are translation invariant)

Remarks on Invariant Distribution

Invariant Distribution

$$\pi = \pi P, \qquad \pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y) \quad \forall y \in \mathcal{X}.$$

- If P is irreducible and there exists a non-negative vector π satisfying $\pi = \pi P$, then $\pi_x > 0$ for all $x \in \mathcal{X}$

  Proof:

  - If P is irreducible, then there exists $N \in \mathbb{N}$ such that every entry of the matrix
  $$A = \sum_{k=1}^{N} P^k$$
  is strictly positive
  - Suppose $\pi_y = 0$ for some $y \in \mathcal{X}$
  By Chapman–Kolmogorov equations, we have
  $$0 = \pi_y = \sum_{x \in \mathcal{X}} \pi_x \left(\frac{1}{N} \sum_{k=1}^{N} P^k(x, y) \right) = \sum_{x \in \mathcal{X}} \pi_x A(x, y),$$
  from which we get that $\pi_x = 0$ for all x **(contradiction!)**

Remarks on Invariant Distribution

Invariant Distribution

$$\pi = \pi P, \qquad \pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y) \quad \forall y \in \mathcal{X}.$$

- Given a TPM P, there may be multiple invariant distributions for P (think of $P = I$)
- **(Important question):** Under what conditions on P does there exist a unique invariant distribution?

On Existence and Uniqueness of Invariant Distribution

Lemma (On Existence and Uniqueness of Invariant Distribution)

An irreducible TPM P has a unique invariant distribution if and only if it is positive recurrent. In this case, the unique invariant distribution $\pi = [\pi_x : x \in \mathcal{X}]$ is given by

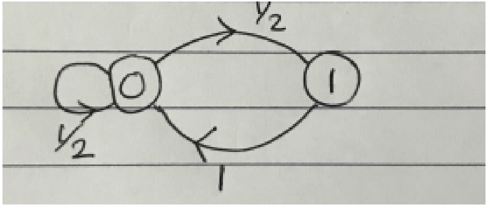
$$\pi_x = \frac{1}{\mu_{xx}} > 0 \qquad \forall\, x \in \mathcal{X}.$$

Summary of Lemma 3:

- Irreducible $+$ positive recurrence $\implies$ unique invariant distribution
- Irreducible $+$ unique stationary distribution $\implies$ positive recurrence

Example

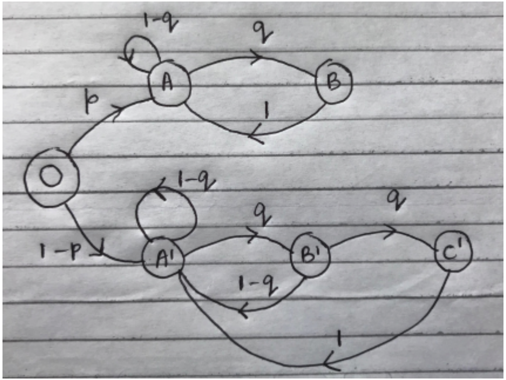
- Consider a DTMC with following transition graph.



1. Is the Markov chain irreducible?
2. Is the Markov chain aperiodic?
3. Classify the states as transient, positive recurrent, or null recurrent.
4. Does a stationary distribution exist for this Markov chain? If so, is it unique?

Example

- Consider a DTMC with the following transition graph.



1. Is this Markov chain irreducible?
2. Does there exist a unique stationary distribution?

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Proof of Lemma 3 (Only If Part)

- Suppose P is irreducible and has a unique PMF π satisfying $\pi = \pi P$
- We must have

$$\pi = \pi \left(\frac{1}{n} \sum_{k=1}^{n} P^k \right) \qquad \forall\, n \in \mathbb{N}.$$

- For all $y \in \mathcal{X}$ and $n \in \mathbb{N}$, we have

$$\pi_y = \sum_{x \in \mathcal{X}} \pi_x \left(\frac{1}{n} \sum_{k=1}^{n} P^k(x, y) \right).$$

- Taking limits as $n \to \infty$ on both sides, we get

$$\pi_y = \sum_{x \in \mathcal{X}} \pi_x \left(\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} P^k(x, y) \right).$$

- If y is transient or null recurrent, then the limit above is equal to 0 (by Lemma 1) for each $x \in \mathcal{X}$
- As a result, we get that $\pi_y = 0$ for all y, contradicting the assumption that π is a PMF!

Proof of Lemma 3 (If Part)

- Suppose P is irreducible and every state is positive recurrent
- We will now construct a vector $\mathbf{v}$ satisfying $\mathbf{v} = \mathbf{v}P$
- Fix an arbitrary $x \in \mathcal{X}$, and define

$$V_x(y) \quad := \quad \sum_{t=1}^{\tau_x^{(1)}-1} \mathbf{1}_{\{X_t = y\}}, \qquad\qquad y \in \mathcal{X}.$$

- Observe that:
  - $V_x(x) = 1.$
  - $\sum_{y \in \mathcal{X}} V_x(y) = \tau_x^{(1)} - 1.$
- Keeping x fixed, define

$$v_x(y) \quad = \quad \mathbb{E}[V_x(y) \mid \{X_1 = x\}], \qquad\qquad y \in \mathcal{X},$$

  and set $\mathbf{v}_x = [v_x(y) : y \in \mathcal{X}]$

Proof of Lemma 3 (If Part)

- Notice that

$$\mathbf{v}_x(x) \;=\; 1, \qquad \mathbf{v}_x(y) \geq 0 \quad \forall\, y, \qquad \sum_{y} \mathbf{v}_x(y) \;=\; \mathbb{E}[\tau_x^{(1)} \mid \{X_1 = x\}] \;=\; \mu_{xx} - 1 < +\infty.$$

- **(Exercise):** $\mathbf{v}_x$ satisfies $\mathbf{v}_x = \mathbf{v}_x P$
- Setting $\pi = \mathbf{v}_x / \mu_{xx}$, we see that π satisfies $\pi = \pi P$
  This proves the existence of invariant distribution for P
- **Uniqueness:** Suppose $\boldsymbol{\alpha} = [\alpha_x : x \in \mathcal{X}]$ is any other invariant distribution satisfying $\boldsymbol{\alpha} = \boldsymbol{\alpha} P$. Then,

$$\alpha_y \;=\; \sum_{x \in \mathcal{X}} \alpha_x \left(\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} P^k(x, y) \right) \overset{\text{Lemma 1}}{=} \sum_{x \in \mathcal{X}} \alpha_x \frac{1}{\mu_{yy} - 1} \;=\; \frac{1}{\mu_{yy} - 1} \;>\; 0.$$